

PH 724 Data Analysis & Scientific Communication Incorporating Use of Generative AI

Part 0: Syllabus Review and Beyond

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Contact

- Instructor: Zhiyang Zhou
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- Lectures
 - Mon 11:30–14:10 @Zilber 110
- Office Hours
 - Mon 16:00–17:00 @Zilber 376

Grading

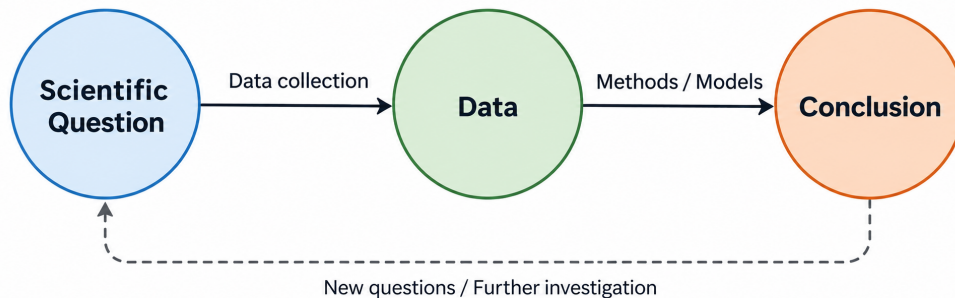
- Reports of Projects 1–4 (60%)
 - Submitting reports (in the format of PDF) via Canvas
 - Attaching both outputs and source codes (if applicable)
 - Including necessary interpretation
 - Organized in a CLEAR and READABLE way
 - Accepting NO late submission
- Final project report (40%)
 - Refer to the final project guideline that will be posted on Canvas
 - Working on any topic of your interest

Materials

- Lecture notes and beyond
 - Posted on Canvas and zhiyanggeezhou.github.io
 - Subject to update **without prior notice**
- Required
 - [Textbook] P. McCullagh. (2022). *Ten Projects in Applied Statistics*. New York: Springer. (Accessible author's copy at www.stat.uchicago.edu/~pmcc/projects.)
 - [P#1] Kass, R. E., Caffo, B. S., Davidian, M., Meng, X. L., Yu, B., & Reid, N. (2016). Ten simple rules for effective statistical practice. *PLoS Computational Biology*, 12(6), e1004961.
 - [P#2] Fernandes-Taylor, S., Hyun, J. K., Reeder, R. N., & Harris, A. H. (2011). Common statistical and research design problems in manuscripts submitted to high-impact medical journals. *BMC Research Notes*, 4(1), 304.
 - [P#3] Barkan, H. (2015). Statistics in clinical research: Important considerations. *Annals of Cardiac Anaesthesia*, 18(1), 74.

- [P#4] Habibzadeh, F. (2017). How to report the results of public health research. *Journal of Public Health and Emergency*, 1(12).
- [P#5] Cao, Y., Chen, R. C., & Katz, A. J. (2024). Why is a small sample size not enough? *The Oncologist*, 29(9), 761-763.
- [P#6] Button, K. S., Ioannidis, J. P., Mokrysz, C., Nosek, B. A., Flint, J., Robinson, E. S., & Munafò, M. R. (2013). Power failure: why small sample size undermines the reliability of neuroscience. *Nature Reviews Neuroscience*, 14(5), 365-376.
- [P#7] Greenland, S., & Finkle, W. D. (1995). A critical look at methods for handling missing covariates in epidemiologic regression analyses. *American Journal of Epidemiology*, 142(12), 1255-1264.
- [P#8] Steyerberg, E. W., Vickers, A. J., Cook, N. R., Gerds, T., Gonen, M., Obuchowski, N., ... & Kattan, M. W. (2010). Assessing the performance of prediction models: a framework for some traditional and novel measures. *Epidemiology (Cambridge, Mass.)*, 21(1), 128.
- [P#9] Mayén, A. L., Guessous, I., Paccaud, F., Stringhini, S., & Marques-Vidal, P. (2016). Educational differences in dietary intake and compliance with dietary recommendations in a Swiss adult population. *International Journal of Public Health*, 61(9), 1059-1067.
- Recommended
 - J. Derr (2000). *Statistical Consulting: A Guide to Effective Communication*. Duxbury Press.
 - J. Cabrera & A. McDougall (2001). *Statistical Consulting*. New York: Springer.
 - G. James, D. Witten, T. Hastie, & R. Tibshirani. (2021). *An Introduction to Statistical Learning: with Applications in R*, 2nd Ed. New York: Springer. (Accessible at www.statlearning.com.)
 - R. Leung (2025). *Leveraging Gen AI for Machine Learning Education in Public Health*. Cham: Springer Nature. (Accessible e-copy via the UWM library.)

Big picture of statistics



AI can support (bio)statistics at each stage

- Scientific question: Synthesize literature, identify knowledge gaps, refine research questions, generate hypotheses.
- Data collection: Design surveys/questionnaires, identify variables, support sampling and recruitment, extract information from EHRs/text/images, automate measurement and data capture.
- Data: Clean and code data, integrate data sources, conduct quality checks, visualize data, support exploratory analysis.
- Methods & Models: Suggest candidate methods, generate code, fit models, check assumptions, perform diagnostics and sensitivity analyses.
- Conclusion: Interpret results, summarize evidence, generate tables/figures, and communicate findings.

Why learn (bio)statistics in this AI era?

AI can calculate faster, write code, fit models, and generate interpretations. But we still need statistical thinking to ask the right questions and judge the answers.

- Ask the right question: Turn a scientific question into something that can be answered with data.
- Understand the data: Know what was measured, how it was collected, and what biases or limitations it contains.
- Choose and evaluate methods: Understand what a method/model assumes and whether it is appropriate.
- Critically evaluate AI output: AI can produce convincing calculations, code, and explanations that are still statistically wrong.
- Make responsible conclusions: Ultimately, researchers must decide what the evidence supports.

What makes this course different?

This is not only a course on methods. It is a course about **statistical practice**:

- How to understand a collaborator's question
- How to choose an analysis strategy
- How to explain results to non-statisticians

Main course components

- Four individual projects
 - Each student will complete a series of individual projects in [Textbook].
 - 3-week cycle for each of the four individual projects

Week	Focus	What you do
Week 1	Introduction (by the instructor)	Understand the scientific question, study design, and data.
Week 2	Preliminary analysis	Clean and explore the data, create visualizations, and develop a plan for formal analysis.
Week 3	Final analysis and presentation	Complete the analysis, answer the scientific question, present your findings, and receive feedback.

- Student-led paper presentations (25-min presentation + 15-min discussion)
 - Each student will present one of the required papers.
 - Each paper presentation should include:
 - * Research or statistical problem
 - * Key statistical lessons
 - * One concrete example
 - * 1–3 discussion questions for audience

Paper assignment

- Please submit top 3 choices by the end of today (Sep 14).
- I will assign as many first choices as possible.
- If multiple competitors want the same paper, I will randomly select one.
- Unassigned students get their second or third choice.
- I will make final adjustments for balance.